

## Project Executable Files

|               |                                                    |
|---------------|----------------------------------------------------|
| Date          | 10/07/2026                                         |
| Name          | Karuru Arjun                                       |
| Project Name  | Voice-Based Concept Understanding Analyser (VBCUA) |
| Maximum Marks | 3 Marks                                            |

### Project Executable Files:

This section contains all executable and deployable files required to run the **Credit Card Approval Prediction** project. The project is built using Python and machine learning libraries to predict whether a credit card application should be approved or rejected based on applicant details. The solution includes the trained ML model, source code, datasets, and documentation for easy deployment and execution.

### Step 1: Submission Checklist

| S.No | Item to Submit                               | Submitted (Yes / No / NA) |
|------|----------------------------------------------|---------------------------|
| 1    | Complete source code (all files and folders) | Yes                       |
| 2    | README / Setup Guide                         | Yes                       |
| 3    | requirements.txt / dependency file           | Yes                       |
| 4    | Database schema / seed files                 | NA                        |
| 5    | Environment Configuration (.env if required) | No                        |
| 6    | Hosted / Deployed Application URL            | No                        |
| 7    | APK / Executable Binary                      | NA                        |
| 8    | Dockerfile / Containerization config         | No                        |

|    |                             |     |
|----|-----------------------------|-----|
| 9  | Test files and test results | Yes |
| 10 | Demo Video / Walkthrough    | Yes |

## Step 2: File / Folder Structure

### VOICE-BASED-CONCEPT-UNDERSTANDING-ANALYSER/

```

|
|— app.py          # Main Streamlit application
|— requirements.txt # Project dependencies
|— README.md       # Project documentation
|— config.py       # Configuration settings
|
|— data/
|   |— reference_topics.json
|   |— sample_audio/
|
|— models/
|   |— whisper_model/
|   |— sentence_bert/
|
|— modules/
|   |— speech_to_text.py
|   |— semantic_analysis.py

```

```

|   ├── audio_analysis.py
|   ├── scoring.py
|   ├── report_generator.py
|   ├── utils.py
|
|── assets/
|   ├── logo.png
|   ├── waveform.png
|
|── reports/
|   ├── generated_reports/
|
|── temp/
|   ├── uploaded_audio/
|
|── screenshots/
|   ├── home_page.png
|   └── result_page.png

```

### Step 3: Deployment / Access Details

| Field                       | Details                                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------|
| Hosted / Deployed URL       | <a href="https://tv4sqbk7ml3pdapcbowcjf.streamlit.app/">https://tv4sqbk7ml3pdapcbowcjf.streamlit.app/</a> |
| Login Credentials           | Not Required                                                                                              |
| Platform / Hosting Provider | Local Machine (Python + Flask)                                                                            |

|                 |                                                                                                                                                                                                           |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Repository Link | <a href="https://github.com/23F21A3110/VBCUA">https://github.com/23F21A3110/VBCUA</a>                                                                                                                     |
| Demo Video Link | <a href="https://1drv.ms/v/c/55933b941b9c6eef/IQC7eH-Dl0jeQaae_pZns1OlAXxok8q9q3DIVz0feLtCy2Q?e=2gFKxe">https://1drv.ms/v/c/55933b941b9c6eef/IQC7eH-Dl0jeQaae_pZns1OlAXxok8q9q3DIVz0feLtCy2Q?e=2gFKxe</a> |

## Step 4: Run Instructions

- Clone or download the VBCUA project repository.
- Open the project folder in VS Code.
- Create a virtual environment (optional).
- Install dependencies:
- `pip install -r requirements.txt`
- Run the Streamlit application:
- `streamlit run app.py`
- Open the browser.
- `http://localhost:8501`
- Upload an audio file (.wav or .mp3).
- Select the reference concept.
- Click **Analyze**.
- View:
- Speech Transcript
- Semantic Similarity Score

## Step 5: Known Issues / Limitations

| S.No | Known Issue                                       | Workaround / Status                      |
|------|---------------------------------------------------|------------------------------------------|
| 1    | Works on local machine only                       | Can be deployed on Render, AWS, or Azure |
| 2    | Requires internet for Whisper model (if API used) | Offline Whisper models can be integrated |
| 3    | Limited predefined concepts                       | More concepts can be added easily        |